

Chapter 3

Business Cycle

Economic Fluctuations

- U.S. economy fluctuates over time.
- State the key regularities in GDP fluctuations.
- Explain the importance of comovement among economic time series.
- State the key properties of comovements among the components of GDP.

Economic Fluctuations

- Discuss why comovements between the price level and real GDP and between the inflation rate and real GDP are important to our understanding of business cycles.
- State the key comovements among labor market variables and real GDP.
- Explain the importance of seasonal adjustment.
- State the key business cycle facts.

Business Cycle Definitions and Regularities

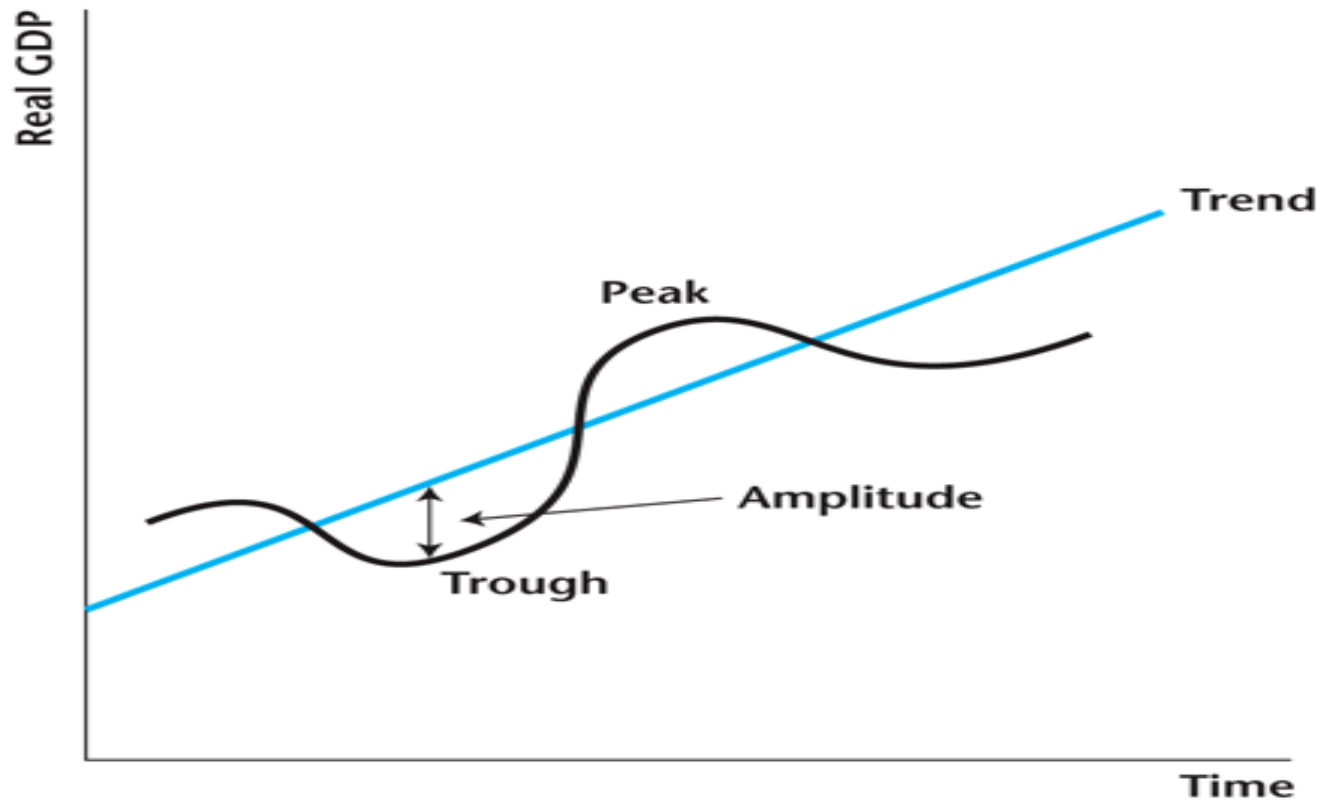
- Recession: Period of declining GDP (relative to trend).
- Boom: Period of increasing GDP (relative to trend).
- Peak: End of boom, beginning of recession.
- Trough: End of recession, beginning of boom.
- No regular amplitude or frequency of fluctuations.
- But there are some regularities or “business cycle facts”:
 - Correlations (comovements)
 - Relative variabilities
 - Phase: leading/coincident/lagging

Business Cycles in US

- US experience: more volatile business cycles pre-war, moderation in fluctuations post-war and especially post 1984.
- Of obvious recent relevance. On July 19, 2021 NBER determined US entered a recession in Feb. 2020, which ended in Apr. 2020.
- Issue of stabilization policy for business cycles: Should Fed cut rates? Should govt. cut taxes? Should government increase spending?

Figure 3.1

Idealized Business Cycles

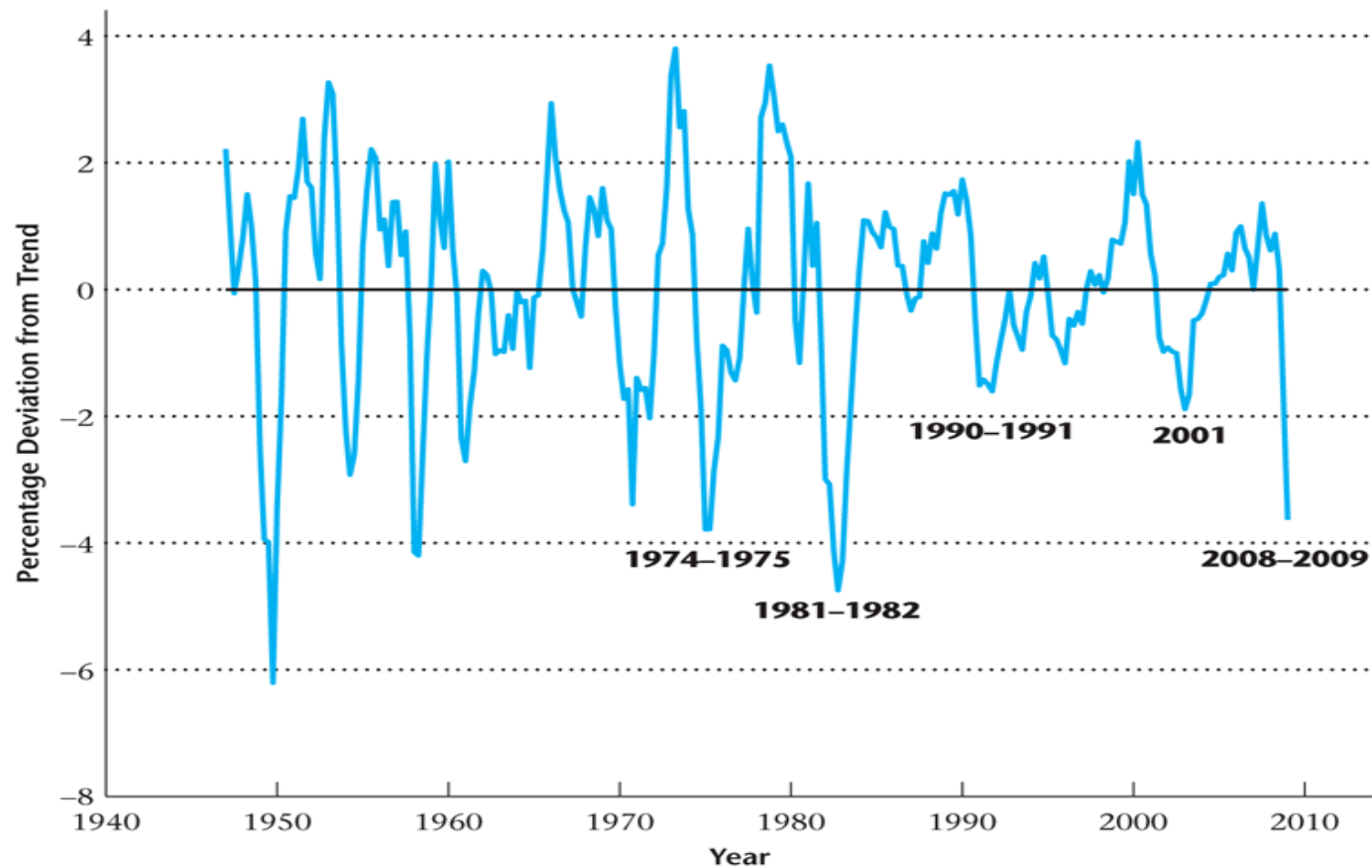


Deviations From Trend in Real GDP are Irregular

- The fluctuations in GDP about trend are quite choppy.
- There is no regularity in the amplitude of fluctuations in real GDP about trend.
- There is no regularity in the frequency of fluctuations in real GDP about trend.

Figure 3.2

Percentage Deviations from Trend in Real GDP from 1947–2009



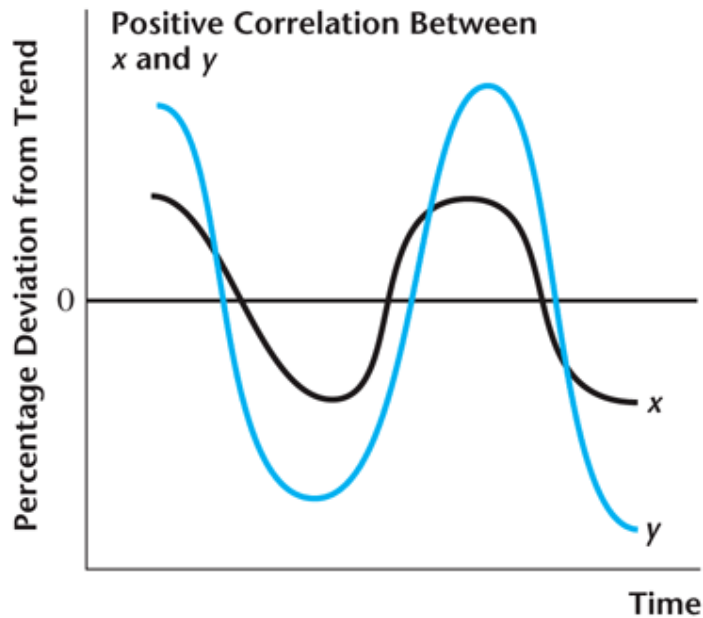
US Business Cycle Expansions and Contractions [1](#)

Contractions (recessions) start at the peak of a business cycle and end at the trough.

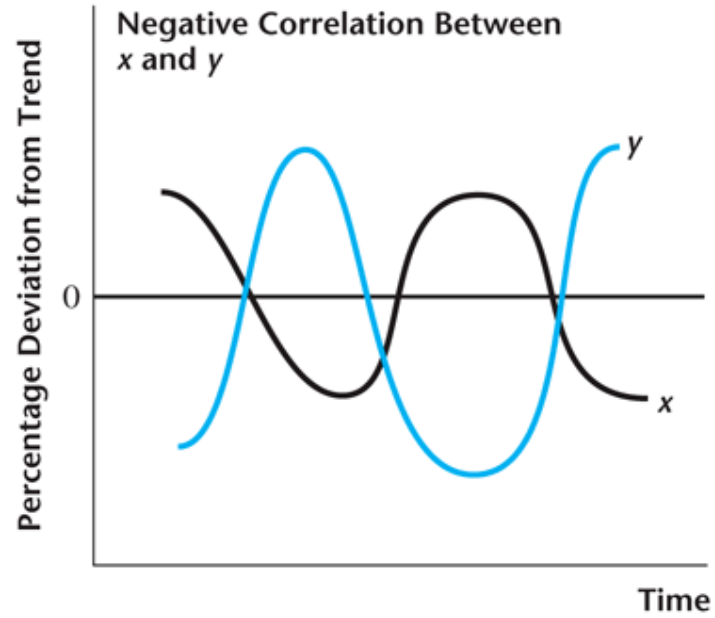
<u>BUSINESS CYCLE REFERENCE DATES</u>		<u>DURATION IN MONTHS</u>			
Peak	Trough	Contraction	Expansion	Cycle	
<i>Quarterly dates are in parentheses</i>		<i>Peak to Trough</i>	<i>Previous trough to this peak</i>	<i>Trough from Previous Trough</i>	<i>Peak from Previous Peak</i>
June 1899(III)	December 1900 (IV)	18	24	42	42
September 1902(IV)	August 1904 (III)	23	21	44	39
May 1907(II)	June 1908 (II)	13	33	46	56
January 1910(I)	January 1912 (IV)	24	19	43	32
January 1913(I)	December 1914 (IV)	23	12	35	36
August 1918(III)	March 1919 (I)	7	44	51	67
January 1920(I)	July 1921 (III)	18	10	28	17
May 1923(II)	July 1924 (III)	14	22	36	40
October 1926(III)	November 1927 (IV)	13	27	40	41
August 1929(III)	March 1933 (I)	43	21	64	34
May 1937(II)	June 1938 (II)	13	50	63	93
February 1945(I)	October 1945 (IV)	8	80	88	93
November 1948(IV)	October 1949 (IV)	11	37	48	45
July 1953(II)	May 1954 (II)	10	45	55	56
August 1957(III)	April 1958 (II)	8	39	47	49
April 1960(II)	February 1961 (I)	10	24	34	32
December 1969(IV)	November 1970 (IV)	11	106	117	116
November 1973(IV)	March 1975 (I)	16	36	52	47
January 1980(I)	July 1980 (III)	6	58	64	74
July 1981(III)	November 1982 (IV)	16	12	28	18
July 1990(III)	March 1991 (I)	8	92	100	108
March 2001 (I)	November 2001 (IV)	8	120	128	128
December 2007 (IV)			73		81

Figure 3.3

Time Series Plots of x and y



(a)



(b)

Figure 3.7

Leading and Lagging Variables

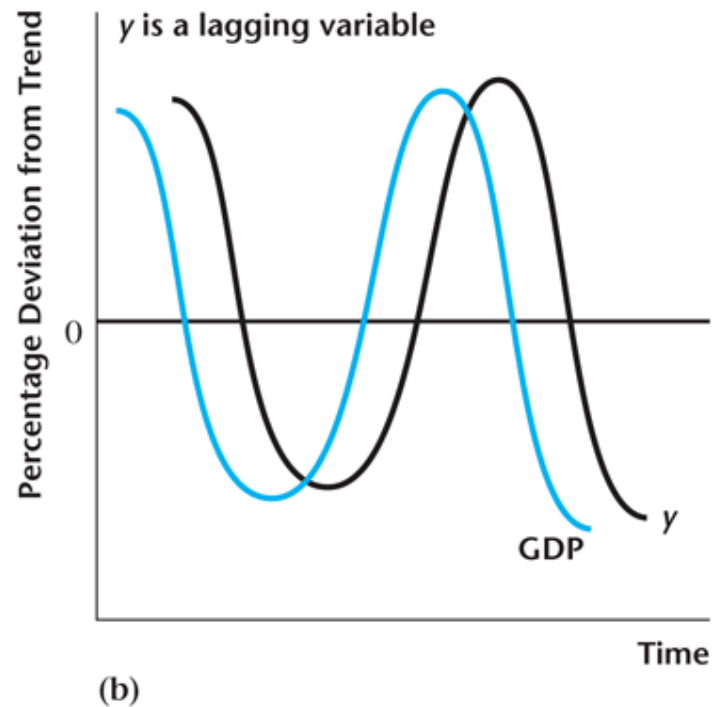
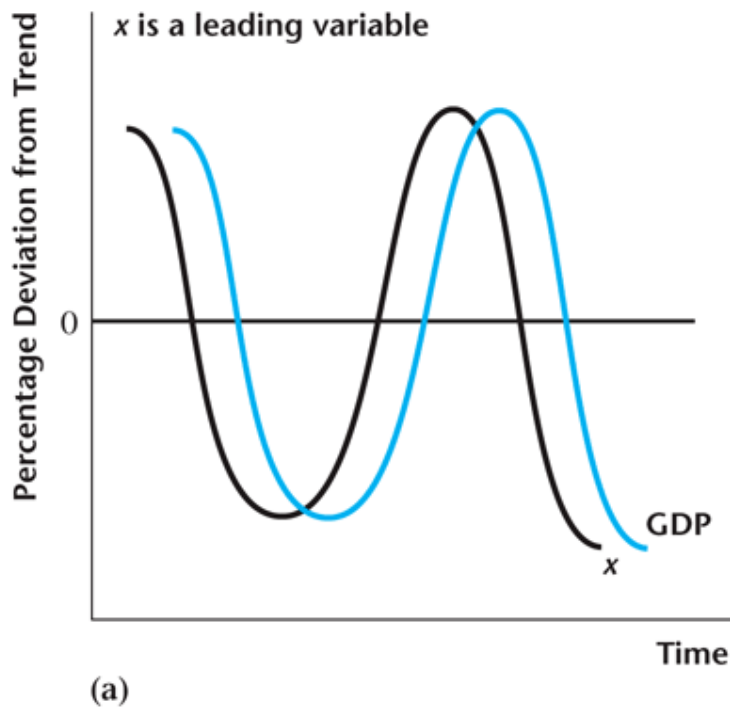
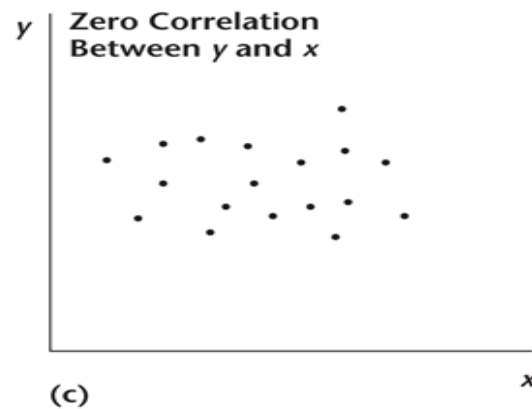
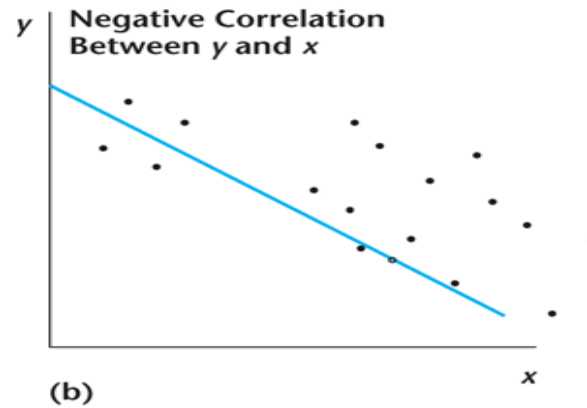
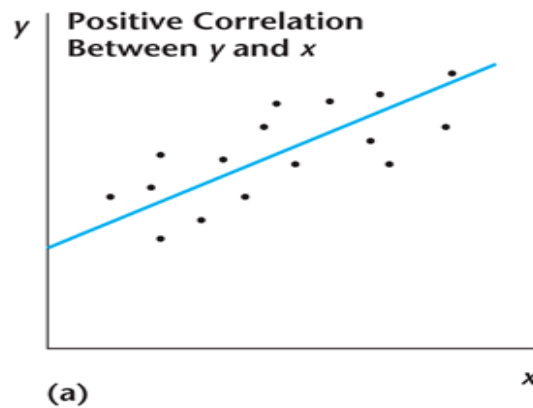


Figure 3.4

Correlations Between Variables y and x



Correlation with Real GDP

- If the deviations from trend in a macroeconomic variable are positively (negatively) correlated with the deviations from trend in real GDP, then that variable is *procyclical* (*countercyclical*).
- If a macroeconomic variable is neither procyclical nor countercyclical, it is *acyclical*.

Figure 3.5

Imports and GDP

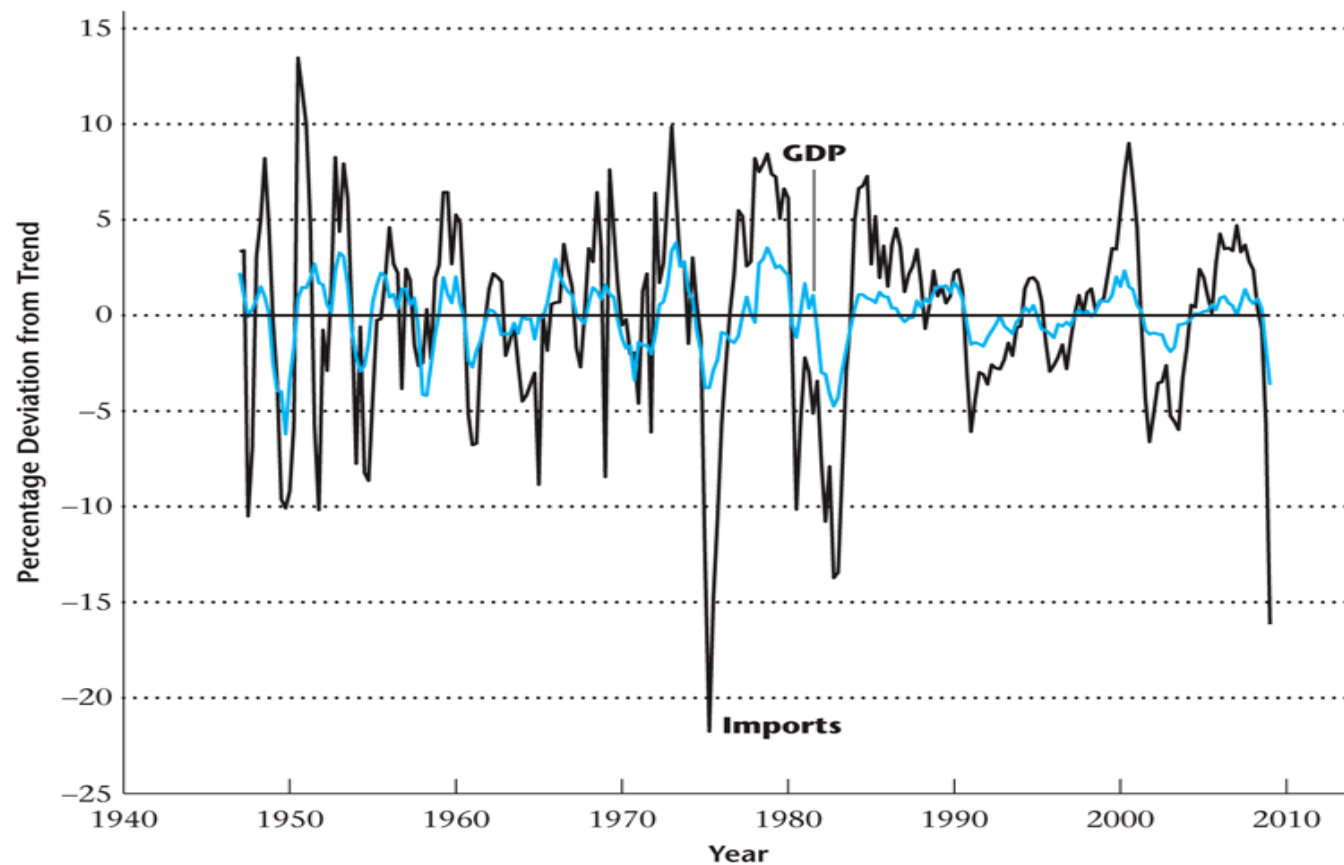


Figure 3.6

Scatter Plot of Imports and GDP

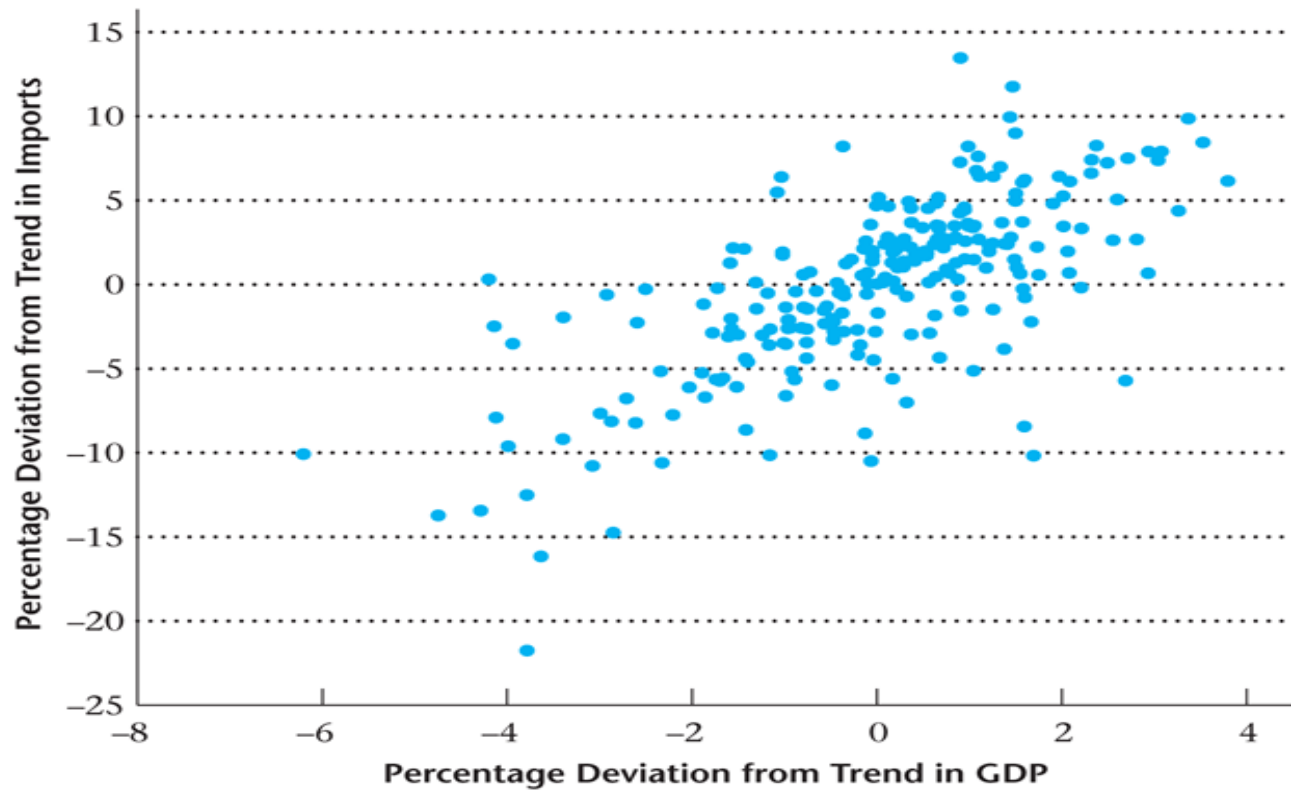
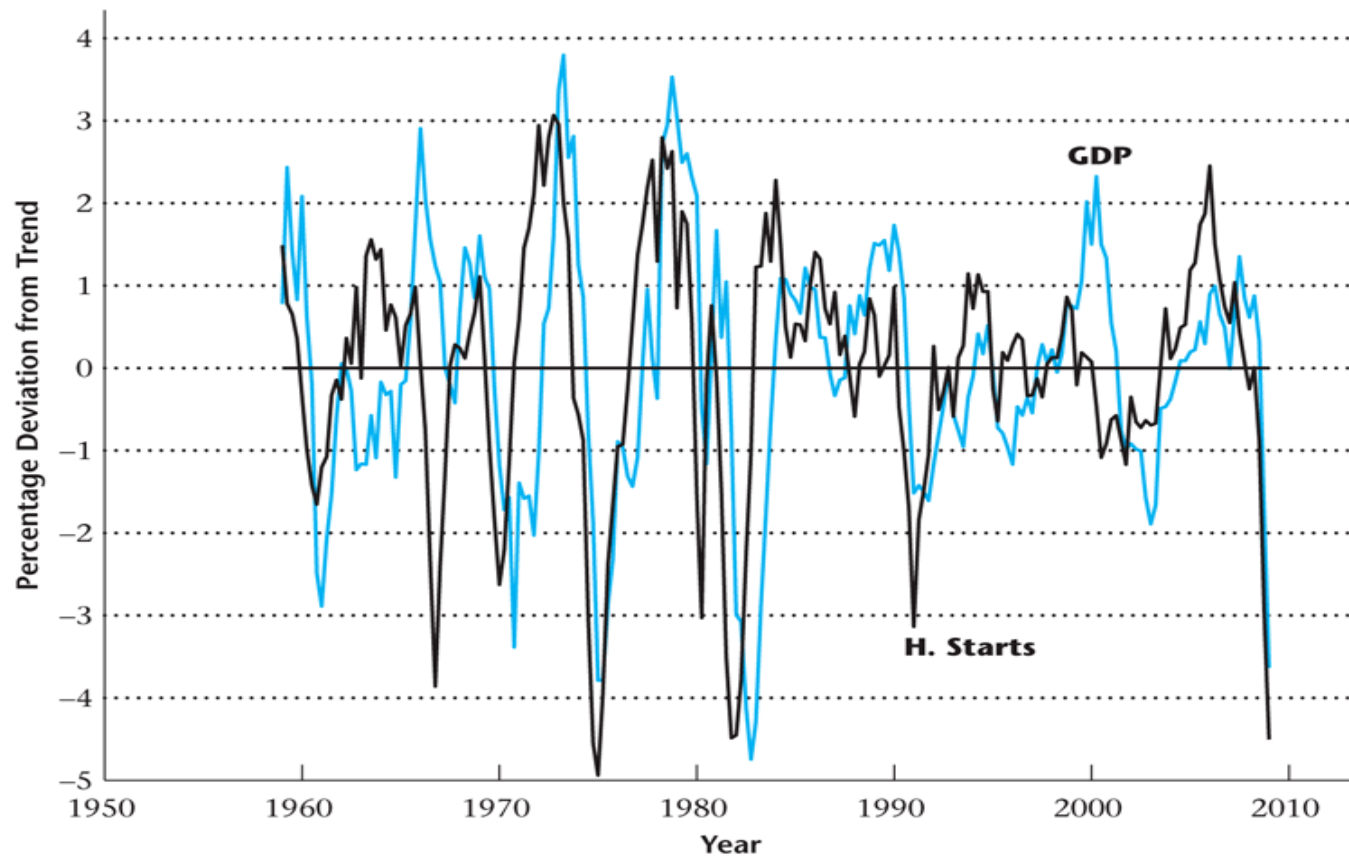


Figure 3.8

Percentage Deviations in Real GDP and Housing Starts, for 1959–2009



Behavior of Key Macroeconomic Variables

- Components of GDP: consumption and investment.
- Nominal variables: price level and money supply.
- Labor market variables: employment, real wage, average labor productivity.

Figure 3.9

Percentage Deviations from Trend in Real Consumption and Real GDP 1947–2009

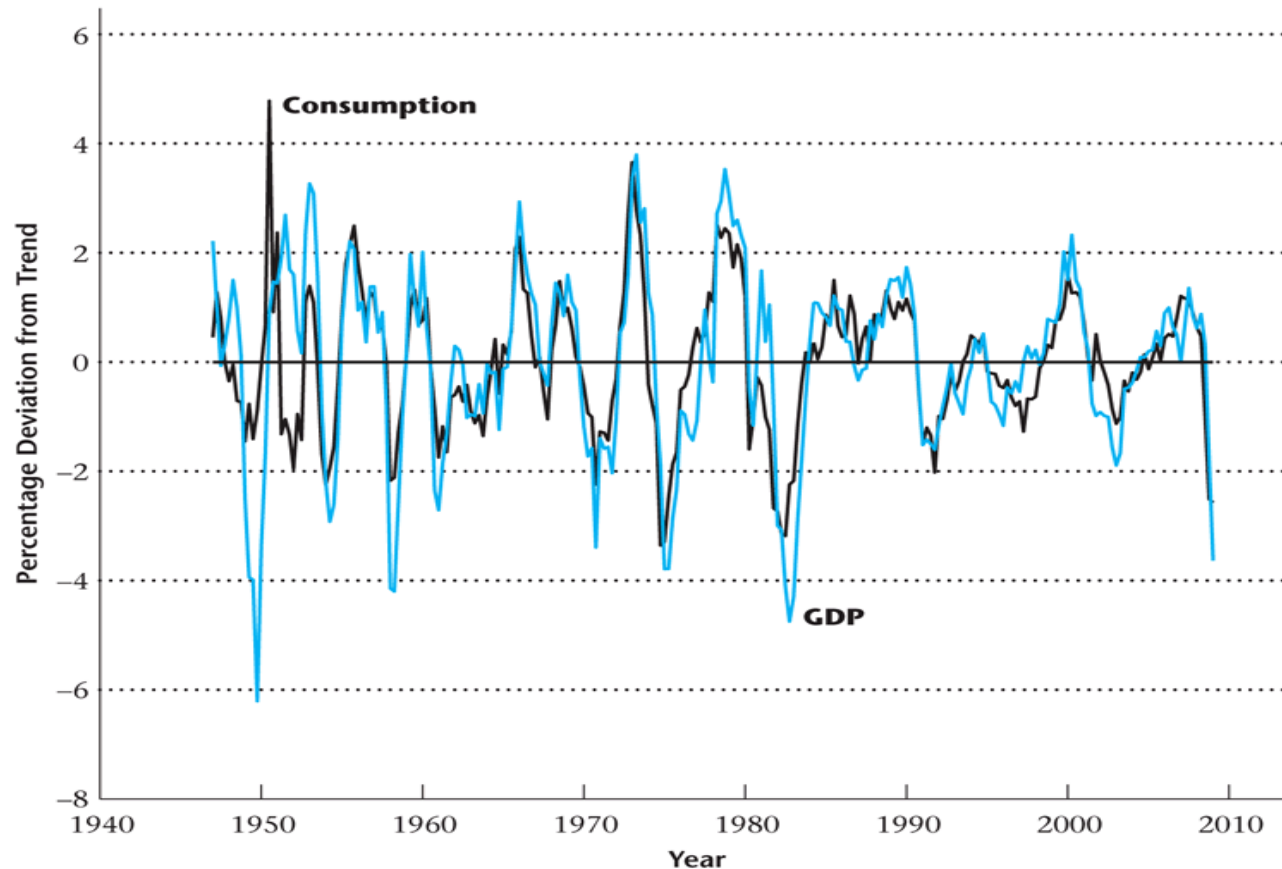


Figure 3.10

Percentage Deviations from Trend in Real Investment and Real GDP

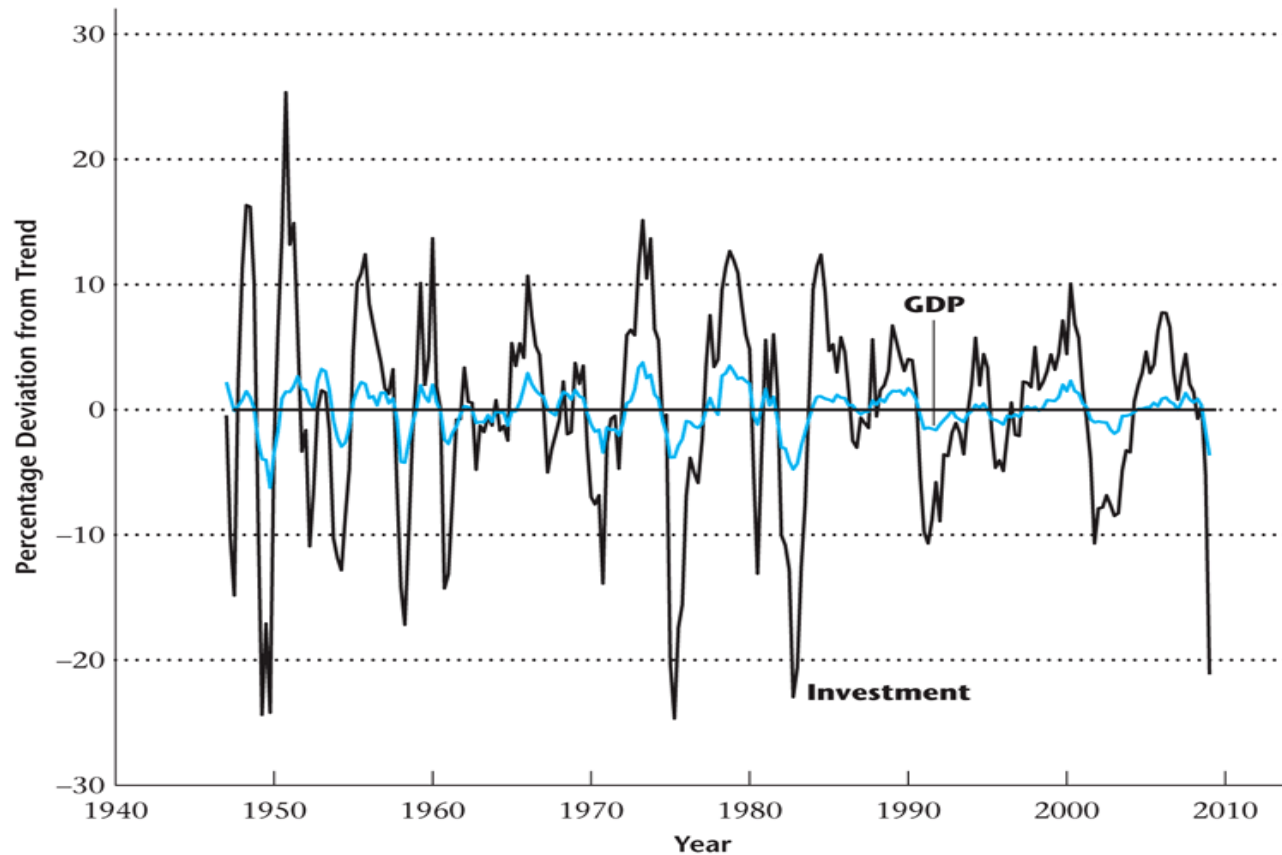


Figure 3.11

Scatter Plot for the Percentage Deviations from Trend in the Price Level and Real GDP

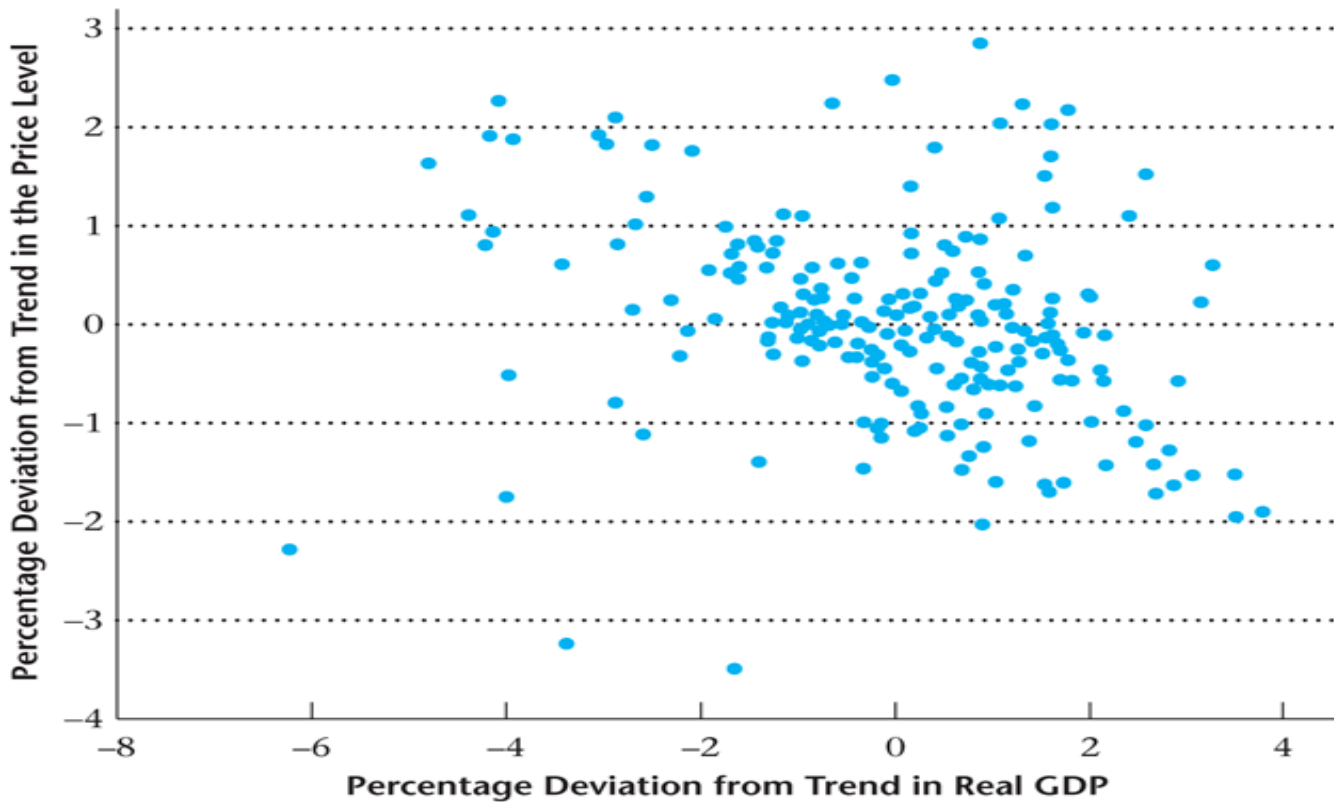


Figure 3.12

Price Level and GDP

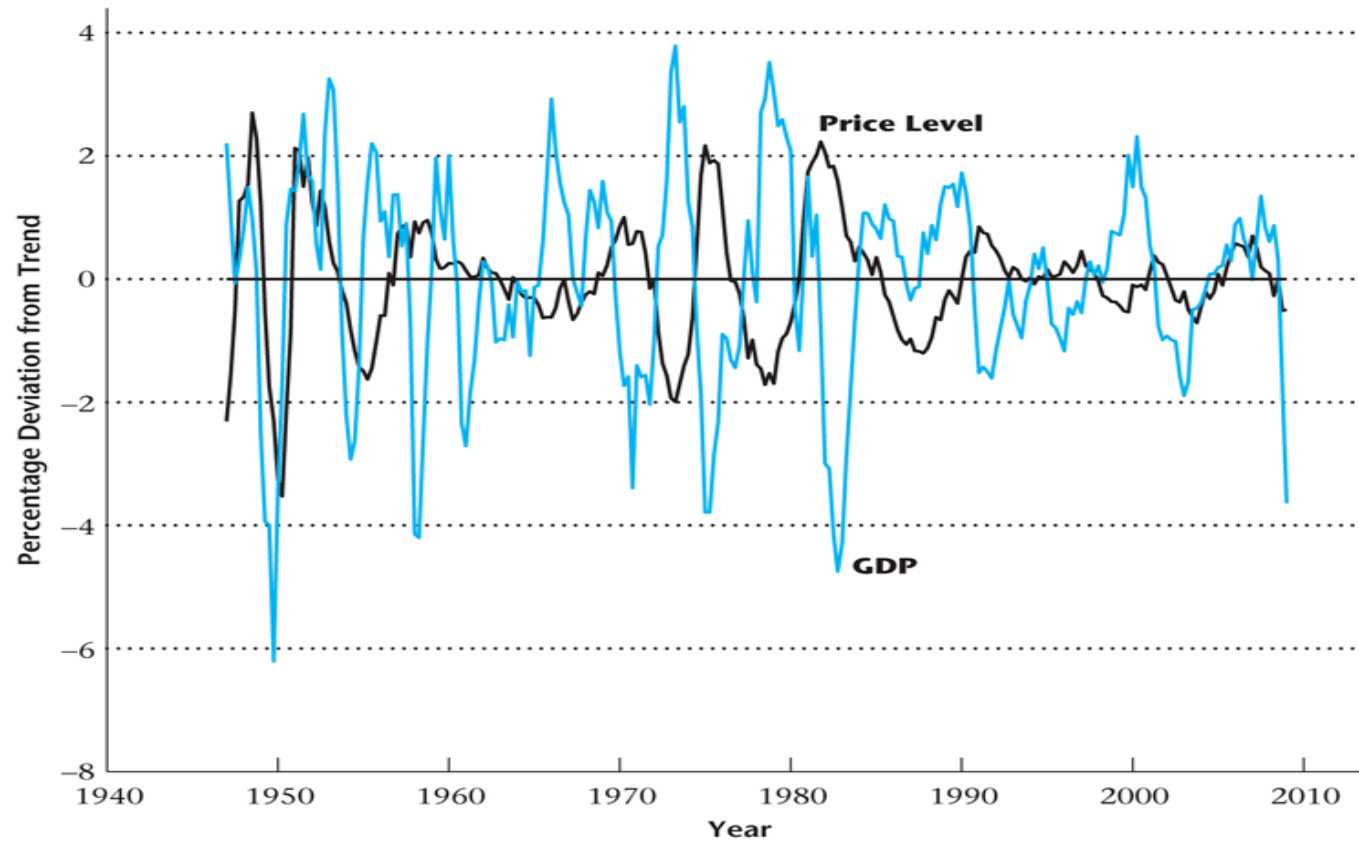


Figure 3.14

Percentage Deviations from Trend in Employment and Real GDP

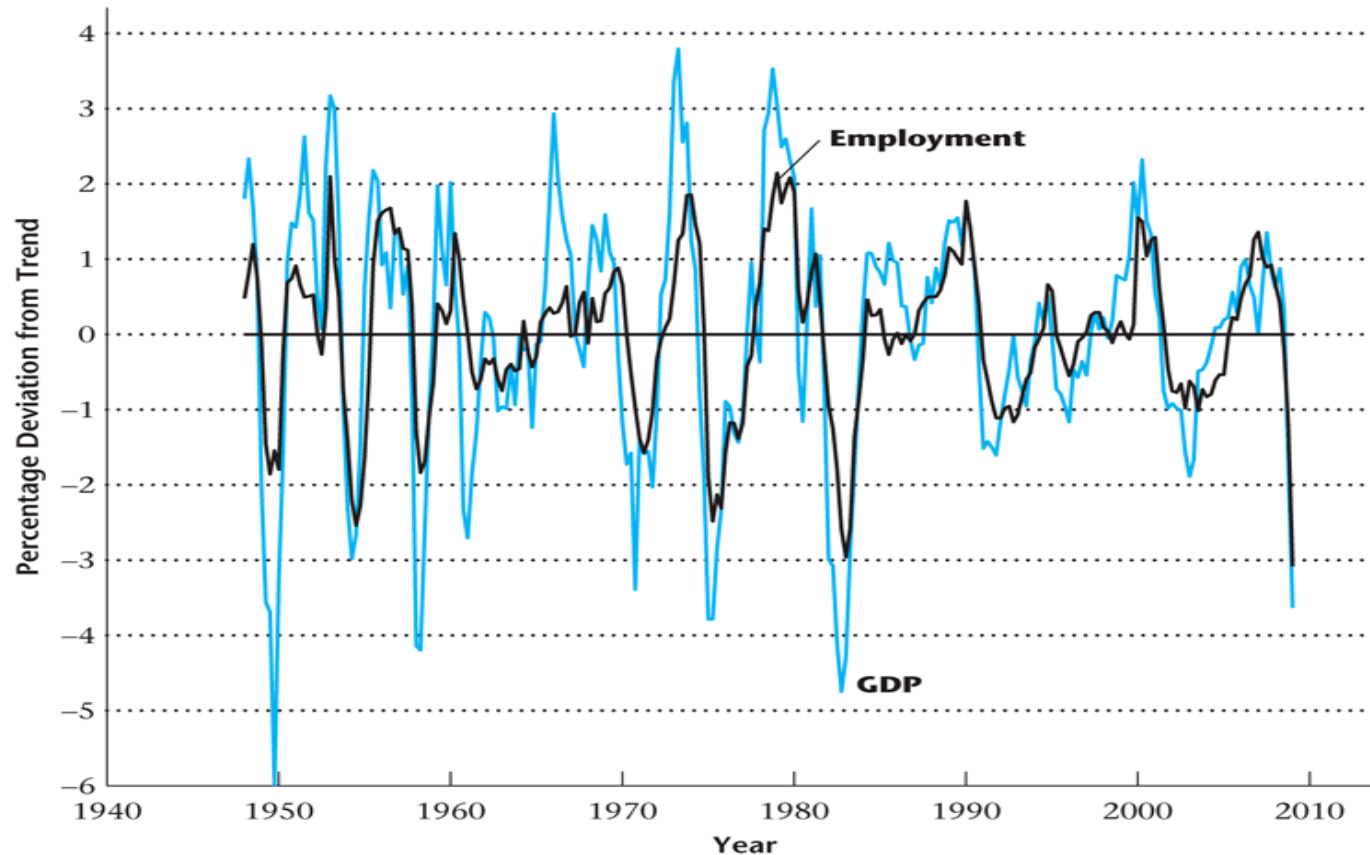


Figure 3.15

Percentage Deviations from Trend in Average Labor Productivity and Real GDP for 1948–2009

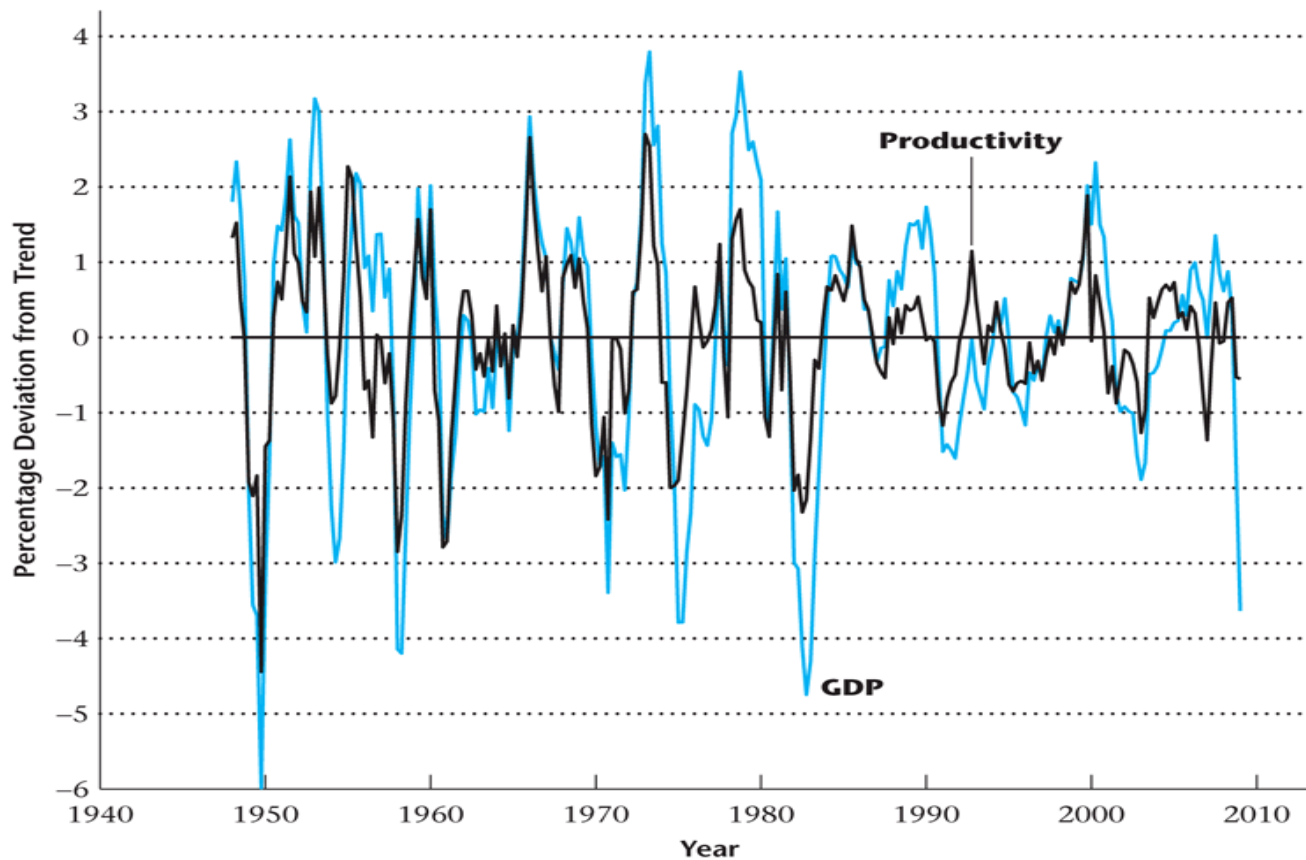


Figure 3.16
Seasonally Adjusted and Unadjusted
Unemployment Rate, 1990-2015

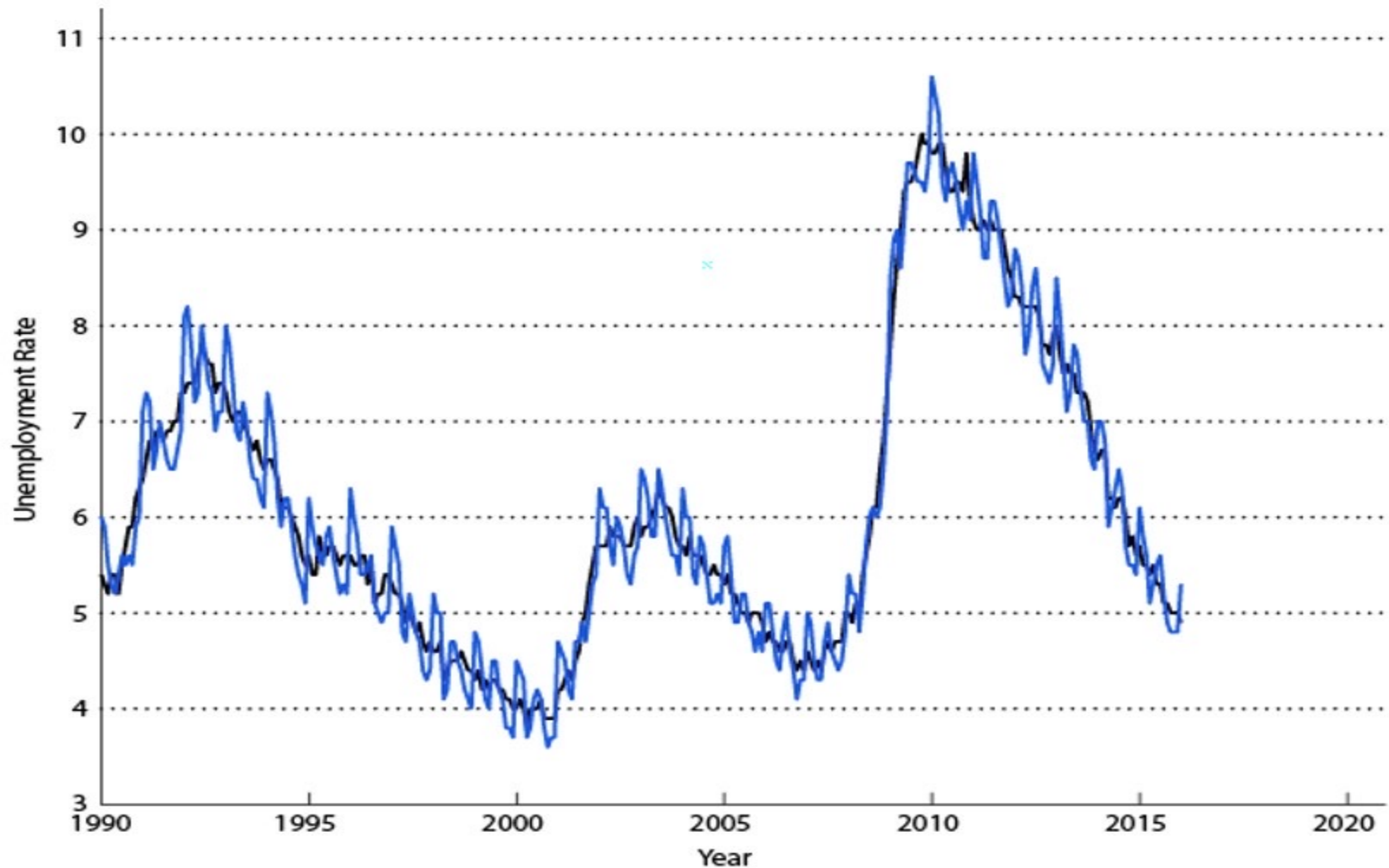


Table 3.1

Correlation Coefficients and Variability of Percentage Deviations from Trend

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Correlation Coefficients and Variability of Percentage Deviations from Trend

	Correlation Coefficient	Standard Deviation (% of S.D. of GDP)
Consumption	0.77	77
Investment	0.80	301
Employment	0.78	65
Average Labor Productivity	0.77	63

Table 3.2

Summary of Business Cycle Facts

Table 3.2

Summary of Business Cycle Facts

	Cyclical	Lead/Lag	Variation Relative to GDP
Consumption	Procyclical	Coincident	Smaller
Investment	Procyclical	Coincident	Larger
Employment	Procyclical	Lagging	Smaller
Real Wage	Procyclical	?	?
Average Labor Productivity	Procyclical	Coincident	Smaller